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Course Code: MLA0401
Course Name: Deep Learning

1.Representation Learning in Deep Learning

Definition

Representation Learning is a Deep Learning technique where the model automatically learns useful features from raw data instead of manually designing features.

Working

1. Input data is provided.
2. Hidden layers automatically learn patterns.
3. Early layers detect simple features.
4. Middle layers combine them.
5. Final layers learn complex representations.
6. Output layer performs prediction.

Traditional ML vs Deep Learning

Traditional ML

Deep Learning

Manual feature extraction Automatic feature learning

Requires domain expertise Learns features automatically

Less effective for images Excellent for images, speech, NLP

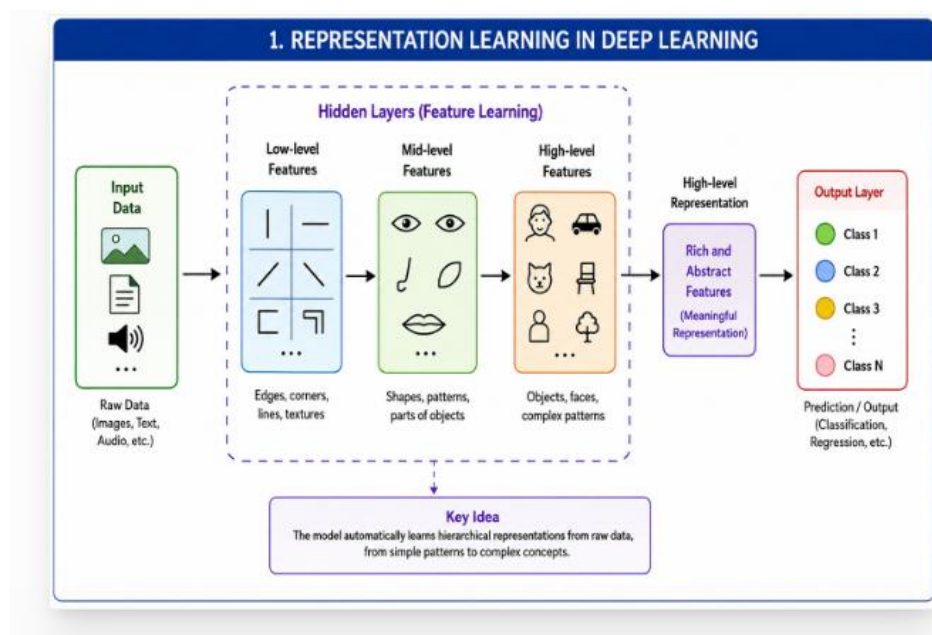
Advantages

- Automatic feature extraction
- Better accuracy

- Handles large datasets
- Learns hierarchical features

Example

Face Recognition, Speech Recognition, Image Classification.



Q2. Width and Depth of Neural Networks

Width: Number of neurons in a hidden layer.

Depth: Number of hidden layers.

Increasing Width:

- Learns more features at one level.
- Faster training.
- Higher memory usage.

Increasing Depth

- Learns hierarchical representations.
- Better accuracy.
- More computation.
- Risk of vanishing gradient.

Comparison

Width

More neurons

Captures broader patterns

Less computation than very deep models

Depth

More layers

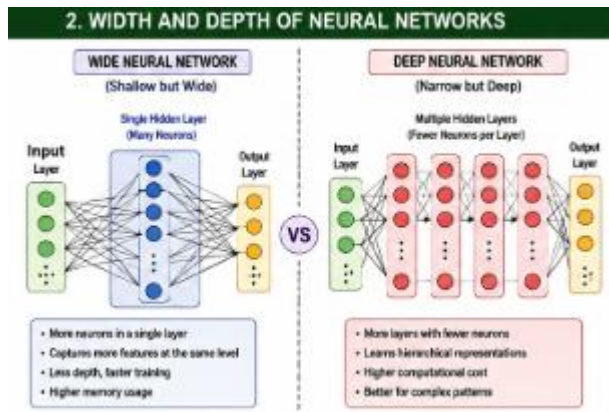
Captures complex patterns

Higher computational cost

Applications

- CNN
- ResNet
- DenseNet

Architecture Diagram



Q3. ReLU, LReLU and ELU Architecture

Advantages

- Faster convergence
- Mean activation near zero

Comparison

ReLU LReLU ELU

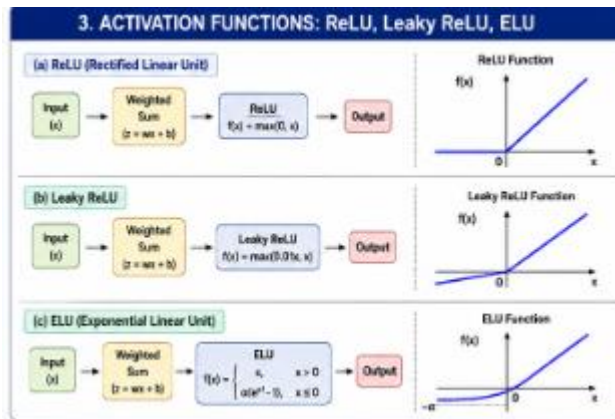
Simple Small negative slope Smooth negative values

Fast Better learning Best convergence

Applications

- CNN
- ResNet
- Image classification

Activation Function Architecture



Q4. RBM and Autoencoder Architecture

RBM: An unsupervised neural network that learns probability distributions.

Working

1. Input enters visible layer.
2. Hidden layer learns features.
3. Reconstructs input.
4. Learns using Contrastive Divergence.

Applications

- Recommendation systems
- Feature learning
- Dimensionality reduction

Autoencoder

Definition

A neural network that compresses and reconstructs data.

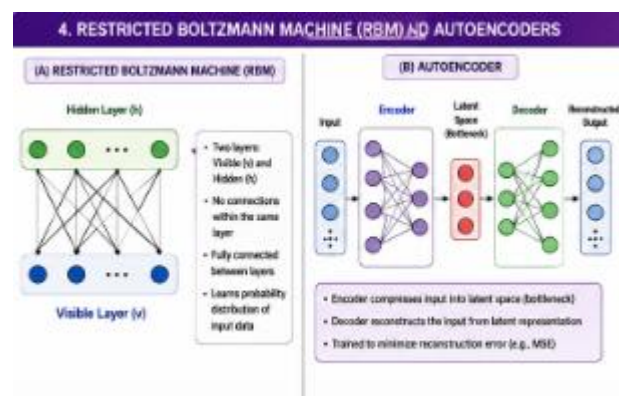
Working

1. Encoder compresses data.
2. Latent space stores important information.
3. Decoder reconstructs original data.
4. Reconstruction error is minimized.

Applications

- Image denoising
- Compression
- Anomaly detection

RBM	Autoencoder
Probabilistic	Deterministic
Energy-based	Encoder-Decoder
Harder to train	Easier to train
Older technique	Widely used



5. Applications of Deep Learning

Explanation

1. Computer Vision

Applications

- Face Recognition
- Object Detection
- Medical Imaging

Examples

- Self-driving cars
- CCTV surveillance

2. Natural Language Processing (NLP)

Applications

- Translation
- Chatbots
- Text summarization
- Sentiment analysis

Examples

- ChatGPT
- Google Translate

3. Healthcare

Applications

- Disease diagnosis
- MRI analysis
- Cancer detection
- Drug discovery

Examples

- X-ray classification
- Medical image analysis

4. Speech Recognition

Applications

- Voice assistants
- Voice typing
- Speaker recognition

Examples

- Siri
- Alexa
- Google Assistant

5. Autonomous Systems

Applications

- Self-driving cars
- Delivery robots
- Drones
- Smart manufacturing

Examples

- Tesla Autopilot
- Warehouse robots

Advantages of Deep Learning

- High accuracy
- Automatic feature extraction
- Learns complex patterns
- Handles big data
- Improves with more training data

